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PALEOMAGNETISM OF MESOZOIC KIMBERLITES IN SOUTHERN AFRICA AND THE CRETACEOUS APPARENT POLAR WANDER CURVE FOR AFRICA

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Abstract. Including those previously published, good paleomagnetic data have been obtained from 48 of 51 sites in 28 separate kimberlites. Radiometric dates (by one or more methods) are available for 17 of these bodies and range from 81 to 145 Ma, with maxima around 90 and 120 Ma. These ages correlate with geochemical contrasts distinguishing what have been designated Group I (younger) and II (older) kimberlites [Smith et al., 1985b]. Stable magnetization appears to be associated with magnetite developed deuterically, after emplacement, but in most bodies, secular variation has not been completely averaged in the remanence acquired. The distribution of "VGPs" calculated is distinctly, but not sequentially, streaked yet correlates broadly with the older and younger petrochemical types. Mean North poles calculated on this basis yield KP1 (N = 14), 81–100 Ma, 64.1°N, 226.1°E, A_{95} = 5.2 and KP2 (N = 10), 113–145 Ma, 47.6°N, 269.9°E, A_{95} = 9.7. With the exception of one undated pipe (Meltonwold), which was not included in the averages, and the melilititic Male fissure at Swartruggens (in Group II average), all the kimberlites sampled are normally magnetized. Including these new data, 10 of 17 published African Cretaceous poles meet acceptability criteria specified, and define an apparent polar wander (APW) curve from ~130 to 75 Ma. The curve appears to consist of two segments, trending latitudinally before ~100 Ma, and meridionally thereafter. The latter trend is evident even in the distribution of the individual KP1 poles, and may be related to the true polar wander speculated by Gordon and Livermore (1987) to have occurred in that interval.

Introduction

Previous paleomagnetic studies of kimberlites in southern Africa have been reported by Jones [1968], McFadden and Jones [1977], Ito et al. [1978], and Hargraves and Onstott [1980] (see Table 1). In view of the high quality of the paleomagnetic data obtained in these studies (usually Fisher's $k \gg 20$), together with the growing number of published radiometric dates for kimberlites (ranging from mid-Proterozoic to Late Cretaceous), a broader sampling of the many known (and mined) pipes in Southern Africa was clearly merited. With generous financial and logistical support from De Beers Consolidated Mines Ltd., an extensive sampling program was initiated in 1981, and continued in 1985. In addition, sets of oriented samples from two pipes were collected by De Beers personnel. The location of the bodies sampled is shown in Figure 1, and the results of the measurements made are reported here.

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Age Distribution of Kimberlites in Southern Africa

Because diamonds have been recovered from the gold-bearing Witwatersrand conglomerates [DuToit, 1954] which are considered to be older than 2.7 Ga, some source of the mineral must have been exposed at the surface by that time. The oldest kimberlites dated, located near Kuruman in the northeastern Cape Province are approximately 1600 Ma [Bristow et al., 1986]. The Premier pipe (the subject of Jones [1968] paleomagnetic study) and others in its vicinity (e.g., National) are approximately 1150 Ma [Allsopp and Kramers, 1977; Allsopp et al., 1988]. Kimberlites giving Cambro-Ordovician (Colossus in Zimbabwe, ~500 Ma) and Permo-Triassic (Jwaneng in Botswana, 250–235 Ma; Dokolwayo in Swaziland, 203 Ma) dates are known, but by far the majority of bodies in southern Africa were emplaced in the Kalahari craton between ~150 and ~85 Ma subsequent to the deposition of the Karoo Supergroup [Allsopp et al., 1988]. That the Karoo strata are essentially flat lying everywhere demonstrates that no structural correction is necessary for the paleomagnetism of these post-Karoo igneous intrusions.

With respect to the younger bodies, Smith [1983] demonstrated that the two varieties of kimberlite long known as "basaltic" and "micaceous," had distinctive petrographic and Sr, Nd, and Pb whole rock isotopic character; he designated these types as Group I and Group II, respectively. It is now also apparent [Smith et al., 1985a] that the two are distinct in age: most Group I kimberlites are about 90 Ma, and are generally less than 115 Ma, whereas Group II kimberlites have emplacement ages varying between 118 and 150 Ma. For the most part, this age difference is clearly reflected in their paleomagnetic vectors (see below).

In anticipation of subsequent discussion as to the paleogeographic meaning of paleomagnetic vectors determined from kimberlites, it should be pointed out that accurate radiometric dating of these bodies is difficult [see Smith et al., 1985a; Allsopp et al., 1988). Mineral phases most frequently utilized for dating purposes are phlogopite (for Rb-Sr and K-Ar) and zircons or rarely, perovskite (for U-Pb). Phlogopite is a conspicuous phase in most kimberlites (especially Group II), either as macrocrysts or as a groundmass constituent. Some micas in kimberlites, however, are xenocrysts from crust or mantle, and inclusion of these in mineral separates used for dating purposes can cause anomalously old ages. Addition of Sr in association with the alteration and metasomatism so prevalent in kimberlites can further complicate the age determination. "Excess" argon may plague K-Ar or $^{40}Ar/^{39}Ar$ age measurements. U-Pb on zircons is a highly regarded method for dating, but zircon is at best a minor accessory, and not all kimberlites contain it. Perovskite of sufficient purity cannot easily be separated. The spectrum of radiometric ages which have been obtained from individual bodies by different

TABLE 1. Published Paleomagnetic Poles From Kimberlites

	Abbrev-iation	No. of Samples	North Poles Lati-tude, °N	Longi-tude, °E	K (k)	A_{95} (α_{95})	Probable Magnetization Age*	Site Location °S	°E
Jones [1968]									
Premier	P	16	51.0	37.0	(26)	(7)	1179±36,Rm 1202±72,Up	-25.7	28.5
McFadden and Jones [1977]									
De Beers	DB	18	56.1	229.9	52	4.8	84Rm,78FT,87A,92Uz	-28.5	24.8
DuToitspan	DP	17	59.9	233.5	35	6.1	84Rm,~83.8Uz,82FT	-28.5	24.8
Wesselton	W	8	57.3	227.6	38	9.1	88Up,90Uz,81Rm	-28.5	24.8
Koffiefontein	KOF	7	54.1	210.6	14	16.9	~90.4Uz	-29.5	25.1
Finsch	F	7	50.6	283.8	71	7.2	119Rm	-28.4	23.5
Hargraves and Onstott [1980]									
Kao	KAO	4	59.6	209.7	(95)	9.5	89±14R U,wr	-29.0	28.6
Letseng la Terae	LET	8	76.0	270.1	(82)	5.7	~94.6 Uz	-29.0	28.9
Lipelaneng	LIP	6	71.7	245.7	(30)	3.8		-28.8	28.3
Marakabei	MA	7	55.4	218.3	(18)	14.4		-28.7	27.9
Pipe 200	P20	6	58.4	235.3	(23)	14.3		-29.0	28.6
Monastery	MON	10	55.9	188.2	(34)	8.4	88±4Rm,90.4Uz	-28.9	27.5
Swartruggens							145±4K;144Rm, 156±13Rm,145Am		
Main	SMA	6	33.9	276.0	(56)	9.0		-25.5	28.5
Changehouse	SC	1	30.6	259.4					
Male†	SML	2	-26.8	111.7	(1099)	7.5			
		6	-29.7	111.5	113	6.3			
National	NAT	11	-21	245	(41)	7.2	1140±16Rm,1180Rm	-25.7	28.5

*Ages are from Allsopp et al. [1988]: K, K-Ar; R, Rb-Sr; A, Ar-Ar; FT, fission track; U, U-Pb; wr, whole rock; z, zircon; p, perovskite; and m, mica.

†An additional four samples were collected from the Swartruggens Male seam in 1987, which, after measurement and inclusion with the original two samples, gave the new result listed here. $n = 6$, $D = 115.6$, $I = 29.9$, $k = 114$, and $\alpha_{95}=6.3^{\circ}$.

techniques, or on repeat determination by the same technique (see Tables 1 and 2) testifies to these difficulties.

The uncertainty in age arising from these problems is emphasized here because, as will be seen below, inconsistencies between the sequence of apparent pole positions and the available radiometric ages may in part be due to uncertainties in the age, independent of the uncertainties associated with the "paleomagnetic" validity of poles calculated from kimberlite remanence.

Sampling and Measurement

The majority of the samples included in this study (see Figure 1), collected both underground and on surface (at least six per site), were obtained with a portable gasoline-powered drill (1-in.-diameter cores). Surface samples were oriented with a sun compass, while those collected underground were oriented by reference to mine-survey reference points using a magnetic compass. In a few cases, hand samples, oriented similarly, were collected and drilled in the laboratory.

All remanence measurements were made on a Schonstedt DSM spinner magnetometer. AF demagnetization (to 130 mT maximum) was performed on a standard two-axis tumbler device, and thermal demagnetization in a Schonstedt apparatus.

Specimens from all samples were AF demagnetized in at least 10 steps up to maximum fields between 70 and 130 mT. For 18 selected samples, thermal demagnetization was performed in 15 to 25 temperature steps up to 680°C. The results in each case were analyzed by orthogonal vector plots [Zijderveld, 1967] and least squares line regression [Kirschvink, 1980]. Acceptable vector segments included at least five demagnetization steps, with mean angular deviation (MAD) of less than 5° (usually less than 2°). Site averages were calculated using Fisher and Bingham statistics [Onstott, 1980]. For analysis of incompletely remagnetized xenoliths, the intersecting great-circle technique of Halls [1978] was employed. The results of all paleomagnetic vector measurements on the kimberlites sampled are presented in Table 2 and Figure 2.

Magnetic Properties

Kimberlite samples from deep open pits or underground mine workings almost always gave excellent results, with near-single-component vectors and broad AF coercivity spectra (see Figures 2 and 3). The stable remanent magnetization (RM) of most Late Cretaceous (Group I) kimberlites is very close to the present-day Earth's field, and any viscous magnetization is essentially indistinguishable from the stable

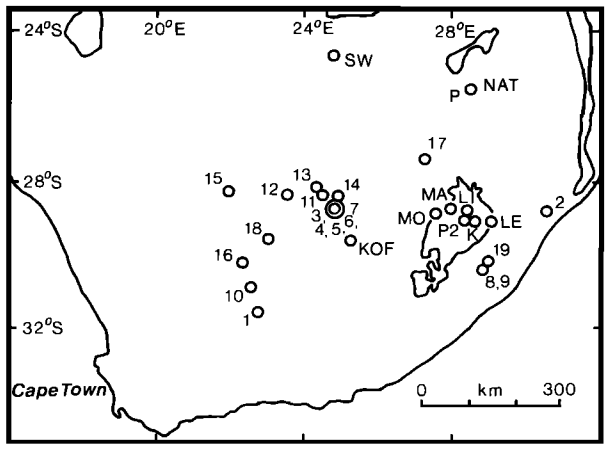


Fig. 1. Map showing location of kimberlites sampled in this (see Table 2) and previous (Table 1) studies: 1, Meltonwold; 2, Emtilomba; 3, DuToitspan; 4, Kamfersdam; 5, Benfontein; 6, Wesselton; 7, DeBeers; 8, Zeekoegat; 9, Abbotsford; 10, Uintjiesberg; 11, Frank Smith; 12, Finsch; 13, Bellsbank; 14, Loxtondal; 15, Makganyeni; 16, Markt, K-15; 17, Voorspoed; 18, Sandrift; and 19, Mzongwana. SW, Swartruggens; P, Premier; NAT, National; KOF, Koffiefontein; MA, Marakabei; MO, Monastery; LI, Lipelaneng; LE, Letseng la Terae; P2, Pipe 200; and K, Kao.

component; as a result, a single linear vector could as well be calculated from the natural remanent magnetization NRM to the peak demagnetizing field (e.g., 4-1-1,2 in Figure 2). Older kimberlites, or surface samples, on the other hand, gave mixed results: in some cases, after removal of a soft viscous component of variable orientation, consistent results were obtained from some or all of the samples collected (e.g., 8-1-1,2 in Figure 2), whereas in others the samples were completely and randomly remagnetized. In the latter case, lightning strikes are suspected, and no useful data could be obtained.

A scalar property measured on all samples used in this study is intensity of NRM (J_n). If petrologically distinct xenoliths are excluded, within-site variation of this parameter is small; on the other hand that between sites (pipes or fissures) can be considerable, with values ranging from >10 to 10^{-2} A/m. The consistency within sites offers support to the presumption made here that other experimentally determined magnetic properties of individual specimens (and petrography of polished thin sections made from them, see below) can reasonably be considered representative of the material sampled at a particular site.

Analysis of J_n (see Figure 4) reveals there is (1) no clear distinction between the magnetic intensity of samples from hypabyssal versus diatreme facies, and (2) a marked distinction between the average intensity of Smith's [1983] Group I versus Group II kimberlites, the former being much stronger than the latter. Although the difference is diminished, Group I kimberlites remain the stronger even after AF demagnetization to 50 mT, but Group II's consistently have a much higher median demagnetization field (MDF) than do Group I's.

The modal abundance of opaque phases in these samples has not been determined quantitatively; but from qualitative observations (see below) there appears to be no variation in abundance comparable to that in J_n. Saturation magnetization (J_S) measurements are available for eight samples only, with Group I kimberlites varying between 1 and 3 A/m, and Group II no more than an order of magnitude less. Curie points of both types (Figure 5) appear to be above 500°C. This is consistent with thermal demagnetization data (Figure 3), which indicates that the blocking temperature of the magnetization was distributed, with 50% of the NRM being unblocked between 450° and 590°C, suggesting magnetite as the carrier. A very small fraction of the remanence persisted above 600°C in some samples, but its orientation was invariably parallel to that of the magnetite component.

All magnetic property data favor the conclusion that magnetite or a magnetite-rich spinel is the predominant magnetic phase in the kimberlites, and carries the NRM.

Petrography and Oxide Mineralogy

Almost all kimberlite samples collected could be classified as either diatreme- or hypabyssal-facies [Clement and Skinner, 1985]. Based on study of polished thin sections from every site, serpentinization and calcitization were variably pervasive, particularly in the diatreme facies; in those samples more severely altered, primary opaque grains could no longer be identified. A variety of primary spinels, in addition to magnesio-ilmenite, have been described from kimberlites, including several of the bodies studied herein [see Boctor and Boyd, 1981, 1982; Shee, 1984; Mitchell, 1986]. Petrographic observations of the kimberlites in this study are generally consistent with Mitchell's [1986] review. Apart from ilmenite, the microphenocrystic oxide grains, when present, are invariably zoned from gray magnesium-aluminum-chrome-rich cores toward browner, more ulvospinel-rich borders. The bulk of the very fine-grained matrix spinels are homogeneous, resembling the outer rims of the zoned microphenocrysts. In size, matrix spinel particles decrease to below the limit of microscopic resolution (1–10 μm). Whether these smallest grains are the product of primary crystallization or are by-products of subsolidus serpentinization or carbonation is not clear. But in either case, they would trend in composition toward purer magnetite [Mitchell, 1986]. Increasing substitution of Mg, Al, Cr, and Ti in magnetite is known to lower the Curie point markedly and progressively [Nishitani, 1981; Schmidbauer, 1983]. Kumar and Bhalla [1984] studied a natural chromite ore and found that it carried a remanance with a blocking temperature of ~350°C. Thus given the composition spectrum of spinel microphenocrysts in kimberlites, they may carry a pure thermoremanent magnetization (TRM) even in diatreme-facies whose temperatures of emplacement may be as low as ~350°C (see below). Later crystallizing spinel phases, which evolve compositionally toward purer magnetite, and therefore have higher Curie temperatures, may carry a post-emplacement thermochemical remanence (TCRM).

On this interpretation, the bulk of the magnetic oxide phases present in kimberlites might be expected to acquire a remanence aligned with the ambient field. The complex zoning and compositional spectrum of microphenocryst spinels may contribute to the broad blocking temperature spectra observed. The high, but distributed AF

TABLE 2. New Paleomagnetic Results From Southern African Kimberlites

Name	Site Coordinates Lati-tude	Longi-tude	Age, Ma	Symbol	No. of Samples	Sampling Location	D	I	k	a_{95}	North Pole Lati-tude °N	Longi-tude °E	K	A_{95}
Meltonwold	31.50S,	22.75E		MW1	4	pipe, trench	160.9	+47.5	101	9.2				
				MW2	4	xenoliths, trench	155.0	+46.3	268	5.6				
				MW3	4	dyke, trench	Scattered, Bipolar							
				MW	8		157.9	+46.9	151	4	-70.7	116.5	133	4.8
Emtilomba dyke	28.25S,	31.54E	81±11,R	EM	6	pit	346.9	-64.0	98	6.8	69.2	238.2	56	9.0
DuToitspan	28.75S,	24,8E	84±3,R	DP1	5	periphery pipe,	358.6	-70.4	207	5.3				
			83.8,U	DP2	6	-576m; core pipe,	351.4	-74.4	348	3.6				
				DP	11	575m	355.1	-72.6	236	3.0	60.4	210.3	86	5.0
				DP*							61.0	222.5		4.2
Kamfersdam	28.7S,	24.7E	86.9,U	K	5	exploratory tunnels	348.6	-66.3	109	7.3	64.1	226.0		10.2
Benfontein	28.81S,	24.84E		B	6	middle sill, quarry	339.5	-63.8	74	7.9	66.5	241.4	38	11.0
Wesselton	28.75S,	24.8E	81±4,R;	W1	6	W.T. sill, 40m sill	339.4	-61.7	381	3.4	68.3	248.6	183	5.0
			90.3,U	W2	7	-40m; pipe D/D5,	337.7	-66.8	172	4.6	63.0	238.3	71	7.2
				W3	6	-805m; pipe D/D11,	343.6	-69.8	82	7.4	61.6	225.9	34	11.6
				W4	3	-805m; dyke D/D11, -805m	343.5	-72.1	199	8.8	59.3	222.6	66	15.3
				W	22		340.3	-67.0	126	2.8	63.9	234.6	55	4.2
De Beers	28.73S,	24.77E	84±3,R;	DB1	5	pre-pipe dyke,-525m	343.6	-67.7	237	5.0				
			92.0,U	DB2	4	pre-pipe dyke,-500m	353.6	-72.1	86	10.0				
				DB	9		347.6	-69.7	128	4.6	63.5	221.4	47	7.6
				DB*							60.0	226.3		6.4
Zeekoegat	30.47S,	28.90E		Z	5	outcrop	27.6	-19.6	14	21.4	57.5		20	17.6
Abbotsford	30.47S,	28.83E		A	4	outcrop	Scattered							
Uintjiesberg	30.83S,	22.53E	100.7±1.4,R	U1	5	outcrop pit, N. end	337.2	-61.2	215	5.2				
				U2	6	outcrop, S. end	332.4	-66.5	197	4.8				
				U	11		334.8	-64.1	174	3.5	65.1	247.2	89	4.9
Frank Smith	28.25S,	24.50E	113.7±1.8,R	FS1	7	FS4, Windsor Block	335.4	-68.7	295	3.5				
				FS2	6	FS2, -160m	337.1	-66.5	494	3.0				
				FS	13		336.2	-67.7	362	2.2	61.3	236.5	146	3.4

TABLE 2. (continued)

Name	Site Coordinates: Latitude	Site Coordinates: Longitude	Age, Ma	Symbol	No. of Samples	Sampling Location	D	I	k	a_{95}	North Pole: Latitude °N	North Pole: Longitude °E	K	A_{95}
Finsch	28.38S,	23.45E	118±2.8,R	F1	7	pipe, -208m	318.6	-51.1	465	2.8	54.3	277.5	290	3.6
				F2	6	pipe, -208m	324.6	-50.2	628	2.7	59.3	279.8	433	3.2
				F3	5	F1 pipe, -244m	308.5	-47.0	390	3.9	45.0	280.7	478	3.5
				F4	5	dyke, -244m	303.8	-51.0	249	4.9	41.8	274.8	231	5.1
				F5	2	precursor dyke, -232m	315.2	-44.3	29	48.5	49.0	291.6	16	68.2
				F6	3	F5 pipe, -232m	307.0	-38.7	116	11.5	42.1	288.7	125	11.1
				F	28		313.8	-48.6	99	2.8				
				F†							50.3	283.1		3.9
Bellsbank	28.1S,	24.4E	120,R	BL1	3	Bobbejaan, brown, -300m	314.0	-66.2	704	4.7				
				BL2	3	Bobbejaan, blue, -300m	302.6	-59.6	49	17.7				
				BL3	5	Main, S-ext, N end -200m	291.8	-67.6	140	6.5				
				BL4	7	Main, S-ext, S end -200m	302.4	-59.8	133	5.3				
				BL	18		301.7	-63.2	95	3.6	41.4	257.2	55	4.7
Loxtondal	28.61S,	24.9E		LX1	7	pipe, -61m	319.2	-56.3	1095	1.8				
				LX2	6	pipe, -121m	314.1	-59.0	530	2.9				
				LX	13		317.0	-57.6	542	1.8	53.2	268.2	343	2.2
Makganyene	28.1S,	22.9E	121±2,R	MK	6	open pit	158.7	60.0	57	9.0	68.5	249.6	28	12.9
Markt, K-15	30.25S,	22.3E	?	K15	6	pits	309.3	-49.5	27	15.2	46.1	278.1		
Voorspoed	27.5S,	27.2E	?	V1	6	open pit	300.7	-62.8	413	3.3				
				V2	6	open pit	300.3	-64.9	137	5.8				
				V	12		300.5	-63.8	217	3.0	40.1	259.1	97	4.4
Sanddrift	29.5S,	29.0E	∿118,R	S	5	?	306.0	-52.6	122	6.9	42.3	274.9		
Mzongwana	30.27S,	29.0E	152.0±3.4,R	MZ	7	outcrop	33.4	-75.0	50	8.6	50.6	186.1	18	14.6
Jwaneng, DK7	24.55S,	24.7E	250±7,R 235±2,U	DK7	6	?	354.1	-60.6	176	5.1	72.0	219.2	102	6.7
Bathlaros	27.3S,	23.3E	1649,R	BT	7	outcrop, pit	164.8	-49.6	57	8.0	-30.0	188.2	47	8.9

*DP and DB are average of this study and McFadden and Jones [1977]; $N = 2$.
†F is average of sites F1, F2, F3, F4; this study and McFadden and Jones [1977]; $N = 5$.

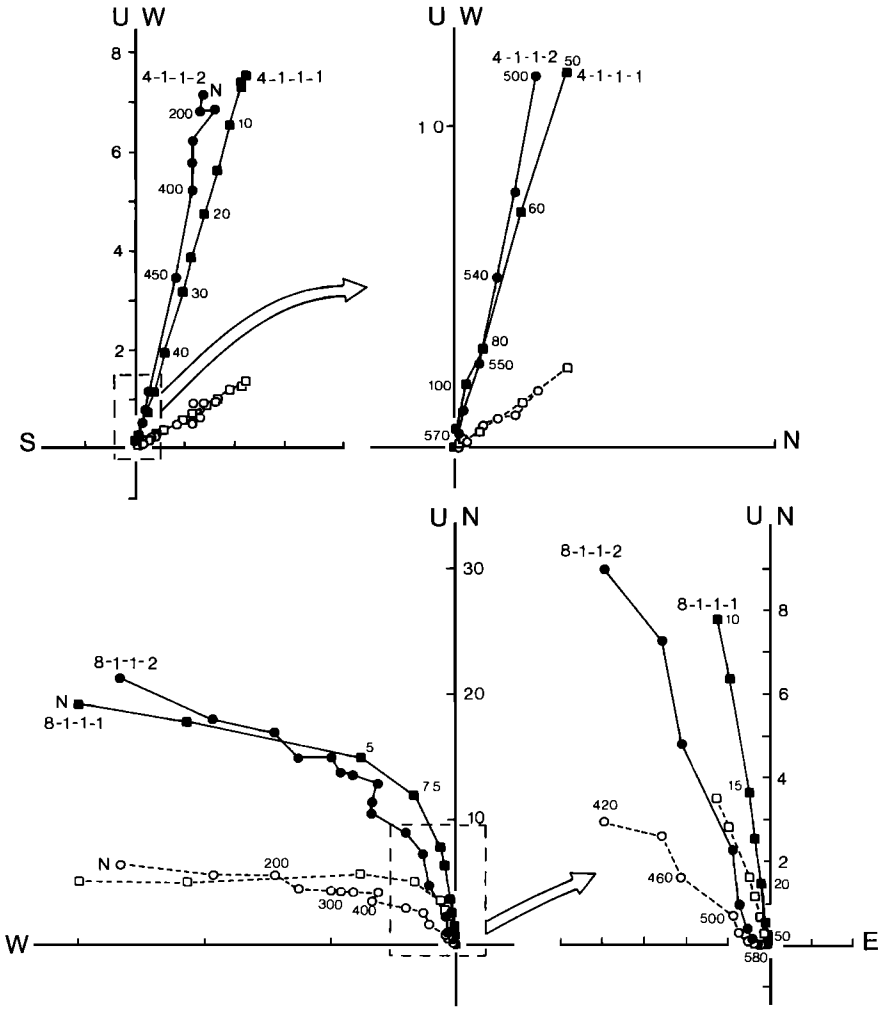


Fig. 2. Orthogonal vector diagrams [Zijdervelt, 1967] illustrating behavior under progressive AF (squares, mT) and thermal (circles, °C) demagnetization of complementary core specimen pairs from represen- tative kimberlite samples. Group I: 4, Wesselton; 8, Benfontein; 17, Uintjiesberg. Group II: 9, Finsch; 12, Bellsbank; 13, Voorspoed. Meltonwold, with reversed remanence, is of unknown age and uncertain petrologic affinity. Open/solid symbols mark declination/inclination; Intensity units are A/m.

coercivity, however, suggests that the bulk of the stable natural magnetization is carried by variably sized single- or pseudosingle-domain magnetite grains, and hence that large multidomain primary spinel grains contribute little to the stable component of the NRM. Such fine-grained magnetite particles may crystallize late, or be produced by alteration of earlier spinels or ilmenites [e.g. see Tompkins and Haggerty, 1985]. They are also a characteristic by-product of the retrograde serpentinization of olivine [Mitchell, 1986, p. 219]. As serpentine is a ubiquitous groundmass mineral in kimberlite (20–50% [Skinner and Clement, 1979]), it can be concluded that to a considerable degree the NRM in kimberlites was acquired thermochemically as a result of the formation of magnetite during the inevitable late-stage hydrothermal alteration associated with volatile streaming. The volatiles involved could be both juvenile and, especially in the diatreme-facies, meteoric water which contributed to the explosive formation of the diatreme by interaction with the rising kimberlite magmas ("hydrovolcanism" [Mitchell, 1986, p. 94]).

Among the broad petrographic distinctions between Group I and II kimberlites, Skinner [1986] reported that in the first, "groundmass spinels and perovskite are typically abundant and relatively coarse-grained. Groundmass ilmenite may be present." In contrast, in Group II, "groundmass spinels and perovskite, if present, are rare and small. Groundmass ilmenite does not occur." These differences are completely in keeping qualitatively with the contrast in intensity of remanence (J_n) between the two types reported in a previous section.

Duration of Magnetization Acquisition in Kimberlites

Mitchell's [1986] review of the emplacement histories of kimberlite pipes illustrates the complexity of the process [see also Clement and Reid, 1986]. Diatremes are created either as a result of the explosive release of primary volatiles [Clement and Reid, 1986] or when kimberlite magma rising up fissures encounters groundwater, leading to explosive eruptions (hydrovolcanism [Mitchell, 1986, p. 96 et

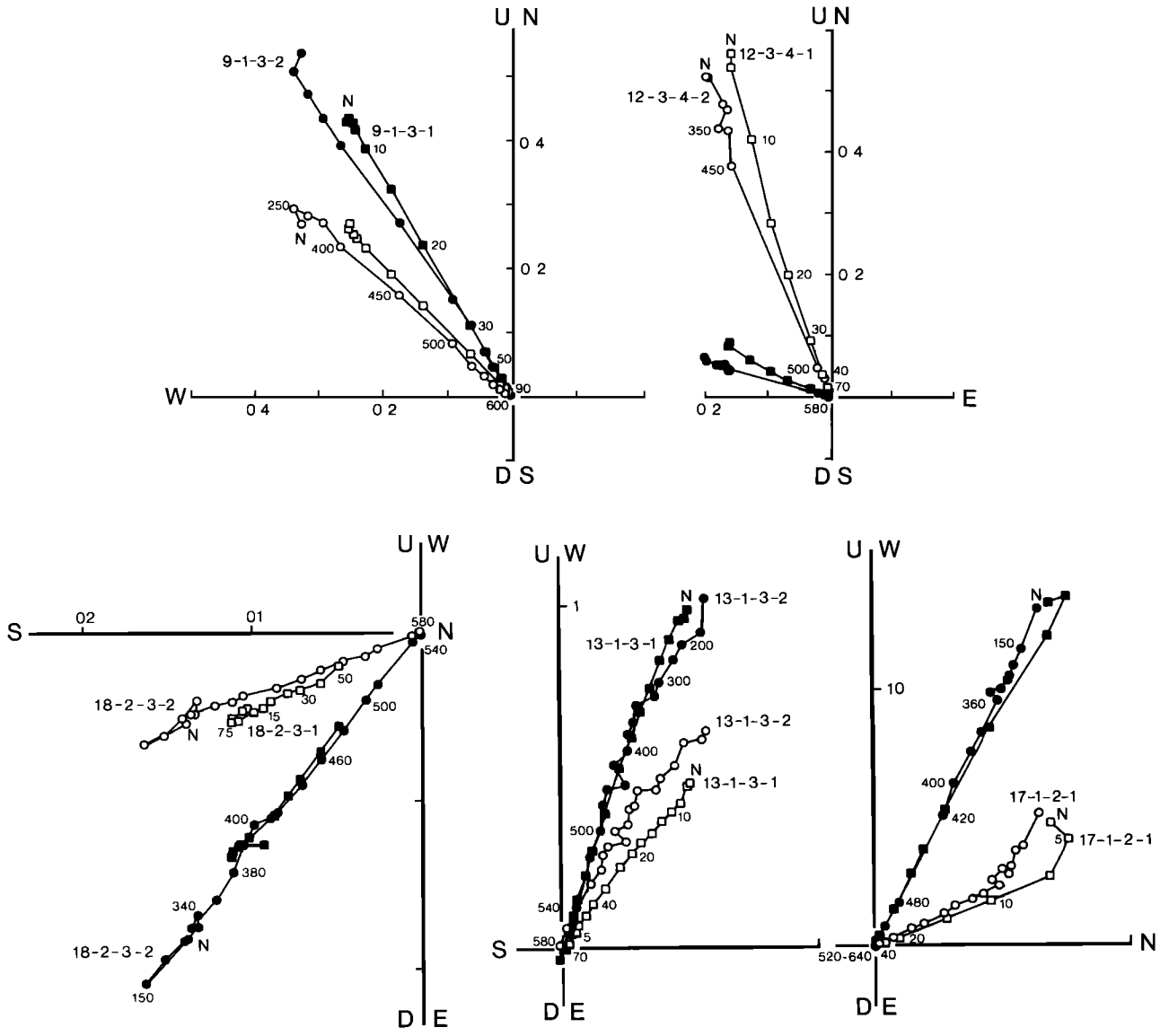


Fig. 2 (continued)

seq.]). Petrographically and texturally distinct phases of "kimberlitic" material can be recognized, both in individual hypabyssal sills and dikes, in the diatreme-facies of the pipes, and in the root zones to pipes (up to 10 at Wesselton, according to Shee, [1984]). A sequence of intrusions from precursor sills and dikes, through pipe emplacement, to, in some cases, post diatreme dikes has often been established. As Mitchell [1986, p. 39], however, states: "The time interval between antecedent dikes and pipe emplacement is unknown." He believes that "The magmatism is likely to be contemporaneous and not resolvable even by the most precise current geochronological techniques." Inasmuch as the smallest age uncertainty reported for the kimberlites dated is at least ±1 Ma, "contemporaneous" may still represent a considerable period when considered with respect to the time scale of secular variation. But if, as asserted above, the NRM of kimberlites is acquired thermochemically in association with post-emplacement cooling and hydrothermal alteration, then the initial temperature at time of intrusion provides at least one boundary condition on how much time may have been involved. Hypabyssal-facies kimberlites originated from magmas with initial temperatures around 1000°C, and primary crystallization may continue down to temperatures approaching 600°C [Mitchell, 1986]. The magnetization of such bodies must include a component of purely TRM carried by the primary spinels. If this is true, however, its direction appears indistinguishable from the TCRM carried by the alteration-related magnetite.

The sudden expansion accompanying the explosive phase of kimberlite pipe emplacement apparently causes a radical drop in temperature. Qualitative evidence of low temperatures of emplacement for the tuffisites of diatreme facies has been inferred from the presence within them of virtually unmetamorphosed xenoliths of bituminous shale, and carbonized wood [Mitchell, 1986, p. 78]. Quantitative estimates, based on the limited thermomagnetic overprints detected in some country rock xenoliths found within four different pipes [McFadden, 1977], indicated that

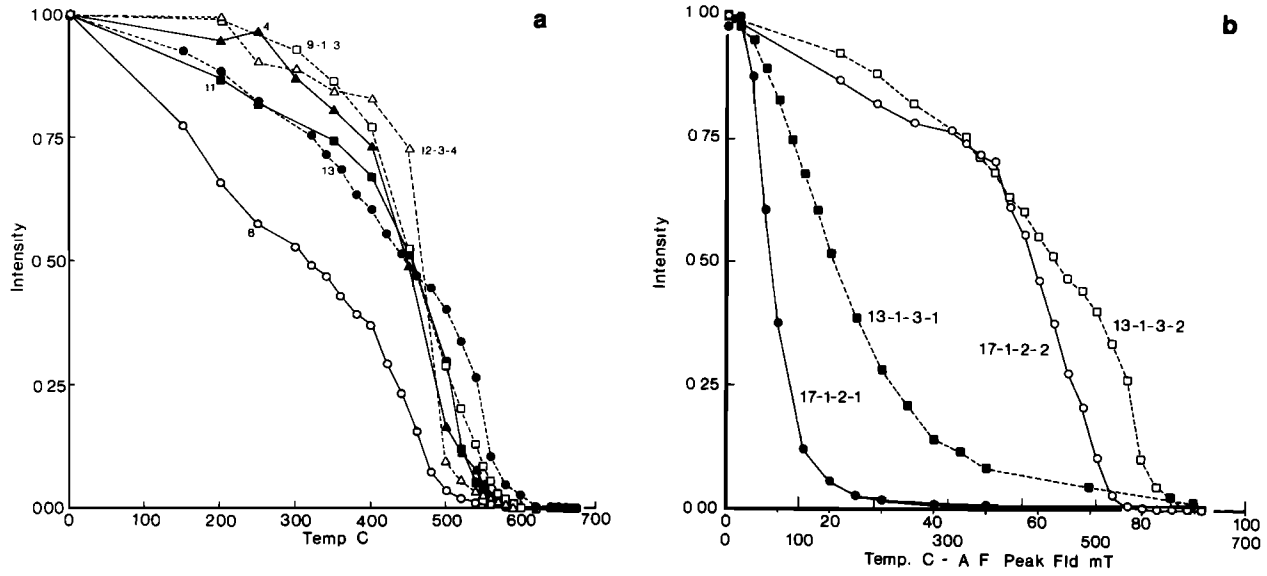


Fig. 3. Normalized intensity (J/J_o) as a function of (a) thermal demagnetization; Group I: 4, Wesselton; 8, Benfontein; 11, Frank Smith; Group II: 9, Finsch; 12, Bellsbank; 13, Voorspoed; (b) AF (solid symbols) and thermal (open symbols) demagnetization of complementary specimens from individual samples from 17, Uintjiesberg (Group I) and 13, Voorspoed (Group II).

the maximum temperature at time of emplacement was 350°C, (and probably closer to 300°C). In the experience of McFadden and Jones [1977], as in this present research, small (<20 cm) xenoliths are usually completely remagnetized to the host kimberlite direction. Petrographic examination of these usually shows evidence of intensive hydrothermal and/or carbonic alteration. It was the rare xenoliths which had escaped complete alteration and in which original magnetization had survived, that were used by McFadden [1977] to infer the low temperature of emplacement of the diatreme-facies kimberlite. That most xenoliths were completely remagnetized to the kimberlite direction despite such low ambient temperatures, testifies to the importance of hydrothermal alteration and thermochemical magnetization in both kimberlite and their included xenoliths in the diatremes.

Kimberlite Poles: Paleomagnetic or Virtual Geomagnetic?

Given the low initial temperature derived for the emplacement of the four pipes they had studied, McFadden and Jones [1977] were led to conclude that each pipe would cool (by conduction) to ambient temperature in times much too short (~400 to 4000 years) to allow complete averaging of secular variation. Consistent with

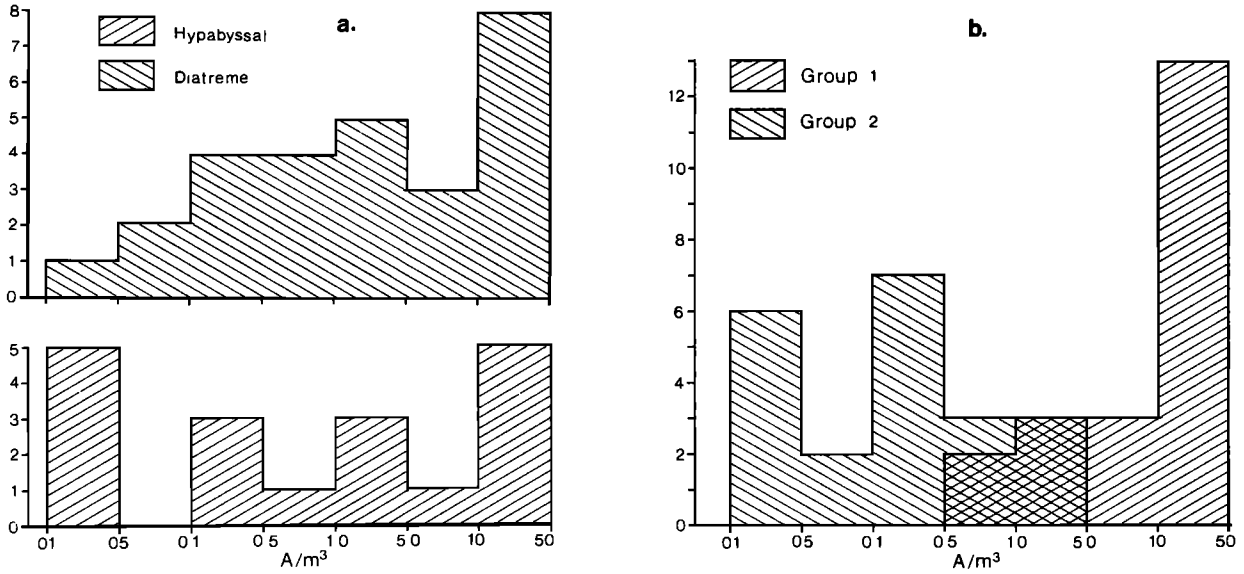


Fig. 4. Histograms comparing average intensity of NRM at sites in (a) hypabyssal versus diatreme-facies kimberlites and (b) Group I versus Group II kimberlites. Note relative similarity on left, and contrast on right.

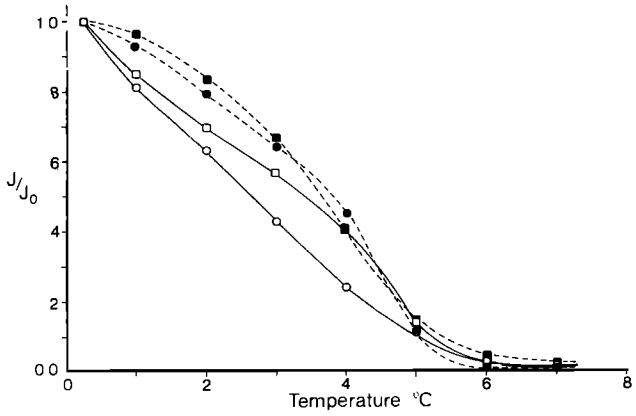


Fig. 5. Normalized magnetization J_o in H=0.5 T. Open circles, Benfontein, middle sill, 0.99 emu/g; Solid circles, Bellsbank, main fissure, 0.31 emu/g. Open squares, Wesselton, Water Tunnel sill, 2.8 emu/g. Solid squares, Uintjiesberg, 2.15 emu/g. Only heating curves (in air) are shown, cooling curves usually being lower.

this, the high precision (Fisher's $k > 35$ for four out of five sites; Table 1) of the results, and the small angular dispersion calculated from it ($\Delta = 81/k^{\frac{1}{2}}$), were generally less than that to be expected due to secular variation at the paleolatitude derived [Brock, 1971]. The three pipes located exactly at Kimberley (De Beers, DuToitspan, Wesselton) had mean directions which were almost identical. Combining all the samples from these three mines gave an angular dispersion that was still too low when compared with that expected if each sample corresponded to an instantaneous magnetization and a time span equivalent to that required to average secular variation was represented in the sampling. McFadden and Jones [1977] therefore concluded that "the De Beers, DuToitspan and Wesselton pipes were all emplaced, cooled, and significant serpentinization of the olivines had terminated in a time less than that required for secular variation to average out." When averaging all five pipes studied, however (De Beers, Dutoitspan, Wesselton, Koffiefontein and Finsch), the statistical parameters were such that they accepted the mean pole calculated as representing a paleomagnetic pole. Subsequent study and radiometric dating, however, have revealed that Finsch (Group II type) is ~30 Ma older than the other pipes (all Group I type), and hence its inclusion in the average was inappropriate.

In the hope of clarifying this question, in this study a deliberate sampling of distinct phases at individual pipes was undertaken. The rationale for this was that if the emplacement of each phase was separated in time, say, by ≥1000 years, and their magnetization acquisition interval was brief, then the NRM of each phase should be distinct. In fact (see Table 2) negligible variation between phases was found. At Wesselton, for example, sampling of four discrete phases (Water Tunnel (precursor) sills, two diatreme phases, and a post pipe dyke) gives nearly identical results (Table 2, W1 to 4). This implies either that the emplacement of the various phases was essentially contemporanous, and all became magnetized in the same 500–1000 years (VGP model), or if the emplacement was temporally distributed, processes causing magnetization continued throughout the entire intrusion episode, and were so prolonged as to average secular variation in each (Paleomagnetic Pole model).

Resolution of this problem has not been achieved, but what is perhaps a significant clue to the magnetization history was obtained from Finsch. Two sites in the main pipe at the 208-m level sampled in 1981 yielded excellent within- and between-site results (see Table 2, F1 and F2). Samples collected in 1985 of kimberlite, and a dyke (or sill) cutting it at the 244-m level again gave excellent agreement (Table 2, F3,4), but the vector is significantly different from the 208-m level average. The 244-m level results, however, were very similar to the Finsch vector reported by McFadden and Jones [1977, Table 1] based on samples they had collected underground at the 348-m level. It is now believed that the present surface exposures at Finsch were no more than ~200 to 300 m below the original surface (De Beers Consolidated Mines Ltd., Geology Department, personal communication, 1987). In which case the 1981 samples represent a depth in the original pipe of ~500 m, those collected in 1985 are from ~550 m, and the McFadden and Jones [1977] collection came from ~650-m depth.

In contrast to this, Hawthorne [1975] has estimated that at Kimberley, the upper 1400 m of the pipes have been eroded. Thus Wesselton samples from the Water-tunnel sills (40 m below present-day surface) and the 805-m level (below surface) in the pipe would then have been ~1450 and 2200 m, respectively, below the original surface; yet their vectors showed little difference.

Kimberlite paleomagnetic vectors thus seem to vary more with (shallow) depth in the pipe than they do from one phase to another at any particular level. The rate of variation with depth seems to decrease with increasing depth below the original surface.

These patterns suggest that (1) the magnetization of kimberlite pipes follows, or continues well beyond, the emplacement of the final intrusive phase. The individual phases, however, may be intruded in rapid succession. (2) Duration of the magnetizing process at any level varies with depth below surface.

It still remains unclear, however, as to exactly how much time is involved in the magnetizing process in the larger kimberlite pipes. In narrow intrusive bodies, as at Swartruggens, in which one of the three fissures (Male) the magnetic polarity is reversed (but not antipodal) relative to that of the others (Hargraves and Onstott [1980], with additions in this study), the magnetization and/or emplacement intervals clearly do not overlap. At depth in the Wesselton pipe, on the other hand, distinctly different facies gave identical results. I judge therefore that the magnetization interval of kimberlite bodies varies as a function of size, and of the depth below the original surface of their present exposure, and that the mean magnetic vector obtained from deep in some large-diameter pipes may indeed approach true paleomagnetic poles.

Kimberlite Poles and Pole Groups

Given that radiometric methods cannot give adequate temporal resolution, an answer to the question of whether kimberlite poles represent true paleomagnetic poles or virtual geomagnetic poles may come from the magnetic data itself. Do the paleomagnetic poles from kimberlite pipes delineate a plausible sequence commensurate with

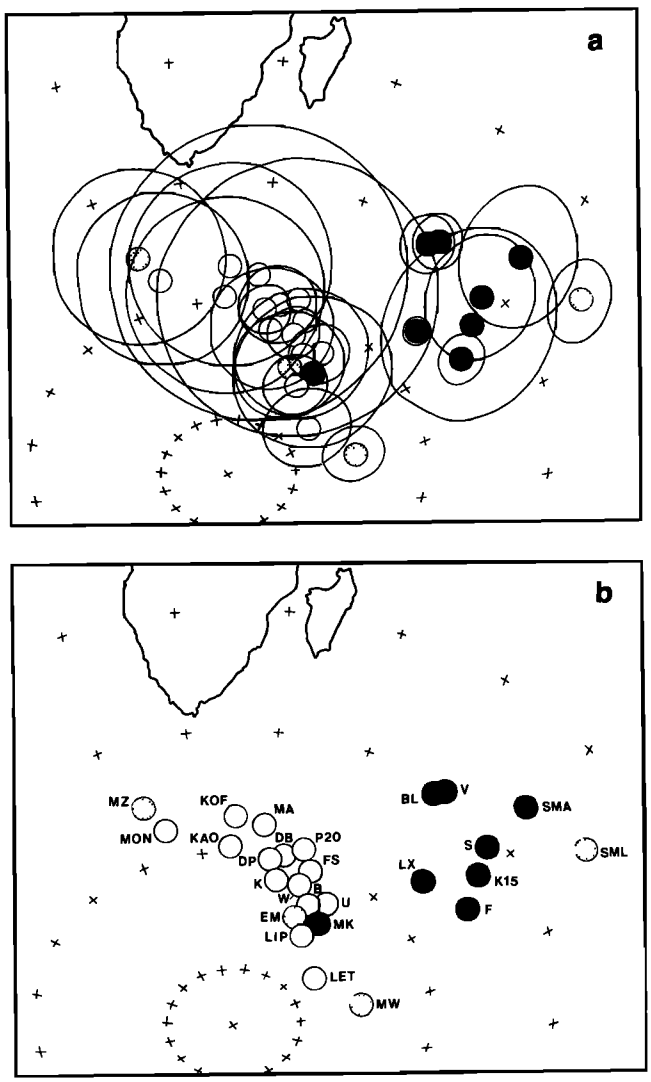


Fig. 6. Equal area plot of poles from all kimberlites (a) with and (b) without α_{95} circle. Pole captions as in Table 2. Solid and open symbols signify Group II and I kimberlites, respectively. Stippled signifies classification uncertainty, or sites are not true kimberlites. Note marked separation of these two groups.

an APW curve for southern Africa? Conversely, for the paleolatitude of southern Africa (~40°S) during the Cretaceous, the angular dispersion to be expected between VGPs in one 10^5- to 10^6-year year interval (during which the APW is considered negligible) is ~16° [Merrill and McElhinny, 1983, Figure 6.17b]. In other words, a sequence of ideal VGPs can be expected to describe a 16° wide swathe which migrates with APW.

The poles calculated for all presumed Mesozoic kimberlites reported in this and previous studies are plotted in Figure 6. Although distinct separation of the paleomagnetic poles of the older (118–142 Ma) petrochemical Group II from those of the younger (~83–114 Ma) Group I kimberlites is apparent, the scatter within each group is considerable, and there is no clear pattern of migration (APW) with age.

This result clearly militates against any assertion that all kimberlites provide true paleomagnetic poles [cf. Hargraves and Onstott, 1980]. On the other hand, the reported difference in radiometric age between pipes which give virtually the same pole is considerable, and it seems highly improbable that the same VGP would be recorded by pipes that are millions of years different in age. Thus if it is argued that identical poles mean identical emplacement and magnetization age, then as pointed out previously, the detailed validity of some of the radiometric ages must be suspect.

Interpretation of these data is further complicated by the fact that the pole distribution is conspicuously streaked (Figure 6). Group I poles in particular delineate an arcuate band less than 10° wide, considerably less than that to be expected from true VGPs. The length of the band suggests that significant APW must have occurred during the time represented; but unfortunately the available ages do not indicate a smooth secular trend.

Of those which have been dated, the oldest Swartruggens (SMA and SML) poles are the "easternmost" of the Group IIs. Mzongwana (MZ) is nominally of equivalent age (~150 Ma) to Swartruggens, but its pole is far removed, being much closer to the Group I swathe. Furthermore Mzongwana, an off craton kimberlite in the East Griqualand field [Nixon et al., 1983], is of Group I petrochemical type! Either its age or its pole is distinctly anomalous. Bellsbank, Sanddrift, and Finsch (BL, S, and F) form a scattered intermediate group (~120 Ma), but the Makganyene (MK) pole, also 120 Ma, coincides with the Group I swathe! If this age is accepted, it should be included with the other Group IIs of this vintage. Although of Group I type, Frank Smith (FS 113 Ma) is closer in age to Group II and perhaps should be included with them.

Poles from seven Group I kimberlites (Tables 2 and 3) having ages between 83 and 92 Ma, form a reasonable group, with three older bodies: Letseng, Uintjiesberg, and Frank Smith (LET, U, and FS in Figure 6*b*) slightly removed. The 81 Ma Emtilomba pole (EM in Figure 6*b*) is very close to this group; but Emtilomba is a melilitite, rather than a kimberlite [Colgan and Allsopp, 1986], so it is not strictly Group I type. Furthermore its age has large errors (±11 Ma). The significance of its pole position is not clear.

Then there are the undated bodies. Given their petrochemical character, and the similarity of their poles, Voorspoed, K-15, and Loxtondal (V, K15, LX in Figure 6*b*) can reasonably be included with the Group IIs. Assignment of the undated Group Is, however, is less certain. Meltonwold (MW in Figure 6*b*) has reverse polarity and hence must be older or younger than the long Cretaceous normal (>118 Ma or <83 Ma [Harland et al., 1982]. Pending age determination, it is thought best to omit it from any average. Benfontein and the undated Lesotho bodies, Marakabei, Pipe 200, and Lipelaneng (B, MA, P20, LIP in Figure 6*b*) all fall in the Group I swathe, and could well be assigned arbitrarily to one or another of Group I subaverages or the overall average. Marakabei (MA) and Pipe 200 (P20) have large α_{95} and perhaps should be excluded on that basis; but by that criterion so should Koffiefontein (KOF in Figure 6*b*). All are included.

Despite their noncircular distribution, various averages of these poles have been calculated and are listed in Table 3. They show no marked variation, and it is not clear which of them is most appropriate to use in defining the African Cretaceous APW curve; for simplicity, the overall Group I and II averages: KP1, ~90 Ma and

TABLE 3. Kimberlite Pole Averages

	Age, Ma	Lati-tude, N	Longi-tude, E	K	A_{95}
Group 1 Kimberlites					
N = 7 (DP,K,W,DB,KOF,MON,KAO	83–92	60.6	215.8	88	6.5
N = 9 (+EM,LET)	81–95	63.9	220.7	51	7.3
N = 10 (+U)	81–100	64.3	223.2	52	6.8
N = 11 (+FS)	81–113	64.0	224.6	56	6.2
N = 15 (+B,MA,P2O,LIP)	81–113(?)	63.9	226.9	62	4.9
KP1 N = 14 (-FS)	81–100(?)	64.1	226.1	59	5.2
Group 2 Kimberlites					
N = 4 (MAK,F,BL,S)	119–121	51.2	268.0	27	17.9
N = 7 (+LX,K15,V)	119–121(?)	49.3	268.1	42	9.4
N = 9 (+SMA +SML)	119–145(?)	45.7	272.4	31	9.4
KP2 N = 10 (+FS)	113–145(?)	47.6	269.9	26	9.7
N = 6 (F,BL,S,+LX,K15,V)	~119–121?	46.0	269.7	80	7.5
N = 7 (F,BL,S,+LX,K15,V,+FS	113–121(?)	48.6	266.3	43	9.3
N = 8 (MAK,F,BL,S,+LX,K15,V,+FS)	113–121?	51.2	265.1	35	9.5
N = 2 (SMA + SML)	145	32.0	283.9	69	30.6
N = 5 (MAK,F,BL,S,FS)	113-121	53.9	262.9	26	15.4

KP2, ~130 Ma have been selected. Note that Frank Smith (FS), 113 Ma, has been included with the Group IIs although it is of Group I type.

Cretaceous Apparent Polar Wander Curve for Africa

Published Cretaceous paleomagnetic data from the African plate are listed in Table 4.

There is considerable evidence to indicate that intraplate deformation occurred in Africa in and along the trend of the Benue rift in association with the opening of the Atlantic [e.g., Burke and Dewey, 1974; Pindell and Dewey, 1982; Castro, 1987; Fairhead, 1988). Inasmuch as the opening of the South Atlantic is considered to have started ~130 Ma ago, any intraplate deformation prior to that would have caused a displacement of Early Cretaceous (and older) poles from southern Africa with respect to those from northern Africa. The magnitude of this displacement, and exactly when it began and terminated, however, is not yet clear. But because the only "acceptable" (see below) Early Cretaceous poles are from southern Africa, and there seems to be little difference between poles <110 Ma years old from the two parts of the continent, the APW path derived in this section strictly should be considered to apply only to southern Africa throughout the Cretaceous and to the entire continent from mid-Cretaceous on.

As May and Butler [1986] emphasized, "the most important factor controlling the accuracy and variability between APW paths is the reliability of the selected data base." The minimum acceptance standards employed by these authors required that a pole must be calculated from the paleomagnetism (after appropriate demagnetization to isolate a stable primary component) in samples from at least 10 sites (multiple samples), the mean having $20<k<150$ (unless site poles are considered to approximate paleomagnetic poles) and $\alpha_{95} \leq 15^{o}$. In addition they required that the age be known to within ±10 m.y. and that there be sufficient discussion of geologic setting to justify any structural corrections. Ideally, the age referred to is the age of magnetization, and if the magnetization measured is indeed "primary," then the stratigraphic (or radiometric?) age of the unit sampled is appropriate. Cryptic remagnetizations that have occurred at times significantly later than primary deposition or crystallization are increasingly recognized, however, and this "age" constraint remains the most subjective of their criteria. To ameliorate this as much as possible, we would add an additional requirement: sufficient discussion of petrography and mineralogy as to support the validity of correlation of the age of the magnetism with the stratigraphic or radiometric age assigned.

We endorse the high standards of May and Butler [1986] and have endeavored to apply them as far as possible to the African data set (Table 4 and the appendix) with the following results:

Nine poles in Table 4 survive the site and statistical requirement (11 if the two Madagascar poles (based on many samples, but for which no site numbers are known) are included. The structural control requirement removes the Sicily (CP) and Morocco (BM, MS1, MS2) poles, if Sicily has been rotated [Besse et al., 1984] and Morocco is considered allochthonous [Westphal et al., 1979]. The age constraint (see the appendix) is much more difficult to impose. The incompatibility between poles from the Nubian sandstones (NS) and the Wadi Natash volcanics (WN1) with which they are intercalated [Schult et al., 1981] strongly suggests that the magnetism of the sediments is secondary, and it is rejected on that basis.

If the number of "sites" required is reduced to five (instead of 10), the additional two (three?) poles entering into consideration are the Bayuda basalts (BA), Lupata Lavas (LU), and Mlanje Massif (ML) (Table 4). According to the authors, the age for the Bayuda basalts is probably an extreme minimum. Pending better age determination, it is excluded. The Mlanje pole was obtained from eight samples only, distributed over 160 m in the rim of the Complex. Whether these results can be accepted as representing eight "independent geomagnetic field readings" [May and Butler, 1986] is not certain. The high k value for the Lupata lavas is disturbing, to say the least, and strictly this pole does not meet the requirement. Although both of these poles need upgrading, they are subjectively included because they may be accurate and appear to fill a gap in the curve.

TABLE 4. Cretaceous Poles From Africa

Name	Symbol	Age* Ma	No. of Sites (Samples)	Latitude, °N	Longitude, °E	K (k)	A_{95} (α_{95})	Reference
Fuerteventura lavas, Canary Islands†	FU	KU,<83	12	68.8	220.8	(99)	(4.4)	1
Bayuda basalts, Sudan	BA	> 74 K	6	55.9	277.8		11.3	2
Nubian ss, Egypt	NS	?	23	81.8	222.7	84	3.3	3
Madagascar dol. & volc.†	MD	~ 75K	(211)	63.5	219.6	42	4.1	4
Capo Passero dikes, volcanics, Sicily	CP	73–82K	37	62	231.0	(38)	(3.9)	5
Kimberlite Pole 1†	KP1	83–101	14	64.1	226.1	59	5.2	This study
Madagascar volc.†	MV	~90K	(170)	69.1	240.0	26	4.9	4
Cenomanian ss, Morocco	MS1	95, strat	5	75	227	(117)	5.5	6
Cenomanian ss, Morocco	MS2	95, strat	7	66.3	227.4	(16)	15.5	7
Wadi Natash Volc., Egypt†	WN1	86–100, K	15	69.3	258.1	21	8.5	3
Wadi Natash Volc., Egypt†	WN2	63–92 K	31	64.6	251.8	(32)	(4.6)	8
Lupata lavas, Moz.†	LU	109–113 K	7	61.8	259.0	(339)	4.0	9
Mlanje Massif†	ML	119–131 K	(8)	60	262	(36)	12	10
Kaoka lavas, Namibia†	KA	113–131K,>130R	40	48.3	266.6	53	(3.1)	11
Kimberlite Pole 2†	KP2	113–142	10	47.6	269.9	26	9.7	This study
Beni Mellal volc., Morocco	BM	170–120 K	13	45	251	26	8	12
Central Atlas intr., Morocco	CA	123–162 K	6	53.0	261.5	10	24	13

References: 1, Storetvedt et al. [1979]; 2, Saradeth et al. [1987]; 3, Schult et al. [1981]; 4, McElhinny and Cowley [1978], 15.050, 15.051; 5, Besse et al. [1984]; 6, Hailwood [1975]; 7, Martin et al. [1978]; 8, Ressetar et al. [1981]; 9, Gough and Opdyke [1963]; 10, Briden [1967]; 11, Gidskehaug et al. [1975]; Allsopp et al. [1984]; 12, Westphal et al. [1979]; and 13, Hailwood and Mitchell [1971].

*K-Ar ages reported in papers published prior to 1979 have been recalculated in terms of the revised decay constants ($\lambda = 5.543 \times 10^{-10}\text{yr}^{-1}$ [see Mankinen and Dalrymple, 1979]).

†Poles meeting acceptability criteria; see text and appendix.

Thus a conservative (yet still somewhat subjective) assessment would select: KA, KP2 ~ 130 Ma; LU, ML ~ 110–120; WN1, WN2 ~ 100 Ma; MV, KP1 ~ 90 Ma; and MD, FU ~ 75 Ma.

A Cretaceous apparent polar wander curve for Africa based on these is illustrated in Figure 7.

Cretaceous Paleomagnetic Euler Pole for Africa?

In a seminal paper, Gordon et al. [1984] argued that because plate motions can be described by approximately constant rotation poles for comparatively long periods (50–100 m.y.), so should the APW paths for the plates for such periods be describable by "Paleomagnetic Euler Poles" (PEPs). This means that for intervals of constant motion of the plates relative to the paleomagnetic pole, the APW paths, ideally, should trace smooth small circles. Major changes in the pattern of plate motions may be manifest by sharp turns ("cusps" or "hairpins") in the APW curves after which a new small circle path defines a new PEP.

Gordon et al. [1984] calculated a PEP to represent the North American Jurassic-Cretaceous APW record. Their analysis revealed, however, that the rate of APW for North America was extremely slow subsequent to 130 Ma. May and Butler [1986] in fact considered it best to calculate only one single average "Cretaceous" pole from these data. North America and Africa were separating throughout the Cretaceous, however, and as their relative rotation pole did not change much, nor was it coincident with the geographic pole [Klitgord and Schouten, 1986], it follows that if North America was in paleomagnetic terms, "quasi-static," then the African APW curve should show considerable motion.

The data compiled in this paper, ranging from 145 to 75 Ma, demonstrate this to be true. For comparison with the African data, in Figure 7 the North American Cretaceous poles selected by Gordon et al. [1984] plus a pole determined more recently by Globerman [1987], have been rotated back to Africa according to the North Atlantic spreading history of Klitgord and Schouten [1986]. There appears to be no significant difference, but the African data, in general being more numerous and of higher quality, reveal considerably more detail. Although a single small circle (PEP path) could be drawn through them, their distribution can be considered to reflect a "cusp" at ~100 Ma, (Wadi Natash poles), with mostly latitudinal motion before, and mostly meridional motion after that time.

Even if this change in APW motion occurring around 100 Ma is real, its significance is not

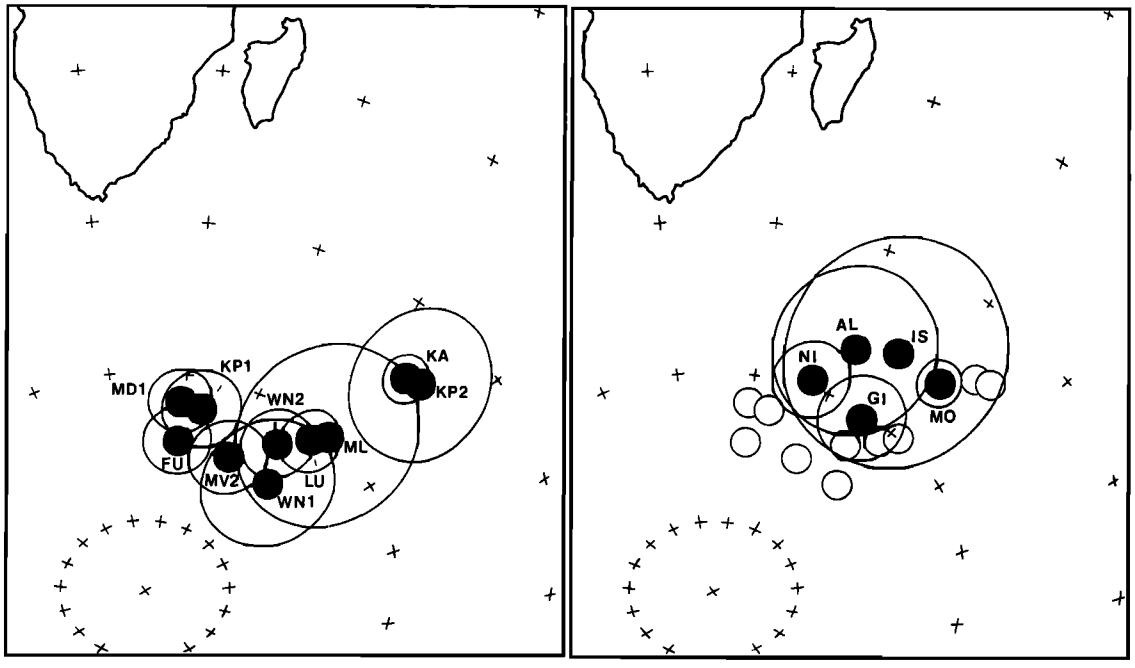


Fig. 7. Cretaceous APW path for Africa depicted by selected poles (Table 4). On right, rotated [Klitgord and Schouten, 1986] North American Cretaceous poles for comparison. Latter poles are as selected by Gordon et al., [1984]. NI, Niobrara, 85–89 Ma; AL, Alkalic intrusives, 98.5 Ma; IS, Isachsen diabase, 109±5 Ma; MO, Monterigian hills, 126±6 Ma; GI, a new 100 Ma pole recently published by Globerman [1987] supersedes AL. The best determined North American poles (MO, GI, and NI) fall very close to the African path, and even AL (A_{95}=11°) and IS (A_{95}=15°) are not significantly different.

presently understood. Because the poles 115 Ma and older are derived from southern Africa only, it is possible that this irregularity is an artifact of the Early Cretaceous deformation referred to earlier [Burke and Dewey, 1974; Pindell and Dewey, 1982]. More and better paleomagnetic data have the potential to provide quantitative solutions to this problem. Certainly after 75 Ma the APW curve for Africa appears to migrate latitudinally (if unsteadily) toward the present pole [Westphal et al., 1986; Livermore et al., 1984], implying another "cusp" in the APW curve at that time. Only between ~100 and 75 Ma is the motion primarily meridional. This irregularity prompts speculation as to whether it might be associated with the true polar wander (TPW) of 14° deemed to have occurred during this interval by Gordon and Livermore [1987]. Such motion, if it occurred relatively rapidly, might even help to explain the anomalous streaking of the poles derived from the numerous penecontem- poraneous ~90 Ma kimberlites (Figure 6).

Research to evaluate further Gordon and Livermore's [1987] assertion of true polar wander [see also Courtillot and Besse, 1987] should be of high priority. As far as Africa is concerned, obvious straightforward contributions could be made by improvements in the dating of the Wadi Natash pole, and further paleomagnetic sampling and dating of the Mlanje Massif. Paleomagnetic and/or radiometric studies of rocks or strata which have already given promising results, but the quality or quantity of which do not meet the rigorous criteria of May and Butler [1986], clearly need upgrading. These constitute attractive "targets of opportunity" whereby the paleomagnetic data base from any continent can significantly be enhanced.

Appendix: Comments on Cretaceous Paleomagnetic Poles from Africa (Table 4)

Central Atlas intrusives (CA). Hailwood and Mitchell [1971]. Six sites, but α_{95} = 24° and K-Ar ages range from 123 to 163 Ma. Unacceptable.

Benni Mellal Volcanics (BM). Westphal et al. [1979]. Good paleomagnetic statistics. Volcanics interlayered with or intruded into Jurassic strata, but K-Ar ages range from 107 to 173 Ma. Possibly two distinct age groups included. Age uncertainty renders pole unacceptable.

Note: Two distinct groupings of radiometric ages were obtained for both CA and BM. It is possible that two distinct periods of igneous activity are represented and poles are similar because of backtracking in APW path [Westphal et al., 1986]. Further study needed to test this possibility. But as Morocco is located in the mobile zone between Africa and Europe, and may be allochthonous, poles from the area should not be used to derive APW path for the craton.

Kaoka lavas (KA) Namibia. Gidskehaug, et al. [1975]. A thorough paleomagnetic study giving a reliable pole. The K-Ar ages reported range from 113 to 131 Ma. Allsopp et al. [1984] have since reported a Rb-Sr isochron age of 132±2.2 for the Messum complex, which is believed to intrude the Kaoka lavas and hence, in the opinion of the authors, provides a reliable minimum age of ~130 Ma for the lavas. Accepted.

Mlanje massif, Malawi (ML). Briden, [1967].

Pole calculated from results obtained from only eight samples distributed over 165 m in the outer syenite ring of the complex, which is several kilometers in diameter. Two K-Ar determinations on biotite and biotite + hornblende concentrates separated from samples of this complex gave ages of 119±6 Ma and 131±6 Ma. It is not certain that these eight samples can be considered to average out secular variation. Marginally acceptable.

Lupata lavas (LU). Gough and Opdyke, [1963]. Single long cores (6–12 feet) at each of four sites, and seven to 10 short cores at each of the other three sites, all distributed over ~1000 m of section. Authors concerned that the small scatter in direction (k=339) means that secular variation was small at this time. McDougall (reported in Gough et al., [1964]) obtained K-Ar ages of 113.3 and 109.4 Ma on anorthoclase separated from the Lupata volcanics. Marginally acceptable.

Wadi Natash volcanics (WN1, WN2). Schult et al. [1981]; Ressetar et al. [1981]. Both represent thorough and complementary paleomagnetic studies. Schult et al. cite published radiometric ages of 100 and 86 m.y. Ressetar et al. [1981] report many new K-Ar results for these rocks showing a range from 63 to 92 Ma with most around 90 Ma. Both accepted.

Cenomanian sandstones, Morocco (MS1, MS2). Hailwood [1975], Martin et al. [1978]. Red sandstones; MS1 from NRM only; MS2 after thermal demagnetization to 450°C; age control is poor. Rejected.

Sicily volcanics (SV). Besse et al. [1984]. A thorough paleomagnetic study incorporating results from previous studies. Measurements on younger rocks, revealed that Sicily had undergone up to 15° counterclockwise rotation in the Pliocene. Westphal et al. [1986], however, concluded that this pole fits on the African curve and suggested either that Sicily had "rotated twice, first clockwise and then anticlockwise, or the ages do not match and the fit of the upper Cretaceous declination is just a coincidence." Whether or not the Cretaceous pole from Sicily shows evidence of rotation relative to Africa depends upon the accuracy of the data from Africa, which this paper is attempting to refine. Given the location of Sicily in the mobile belt between Africa and Europe, however, data from Sicily cannot be used in constructing the cratonal APW path. Omitted.

Madagascar Volcanics (MV1, MV2). These data, based on an unpublished dissertation by C. Andriamirado (1971), are reported in McElhinny and Cowley [1978]. Many samples were studied, apparently, but details as to the number of sites, etc., is not available. Accompanying age data indicated two groups at ~75 and 90 Ma, and McElhinny and Cowley [1978] report average poles for these two time periods. As Madagascar has been attached to the African plate in its present position since ~105 Ma [Rabinowitz et al., 1983; Tauxe et al., 1983]; these poles are applicable to the craton. Accepted.

Nubian sandstones (NS). Schult et al. [1981]. A thorough study, but intercalated Wadi Natash volcanics give completely different direction. This strongly suggests that these rocks are remagnetized in recent Earth's field direction. Rejected.

Bayuda basalts. Saradeth et al., [1987]. Sound paleomagnetic study, but results from only six sites. Although an average of seven samples was taken at each site, it is not known how many separate flows were represented, and hence secular variation may not be averaged in this pole. The only published age data (K-Ar whole rock) indicated ages 62–74 Ma; a new radiometric study is in progress. Provisionally rejected.

Fuerteventura lava. Storetvedt et al. [1979]. A thorough paleomagnetic study, and the isolation of primary magnetization is well documented, indicating (on stratigraphic grounds) acquisition in uppermost Cretaceous. Miocene volcanic activity apparently caused argon loss from these rocks, so that they give much younger K-Ar ages. It seems that the structural disturbance of the island had ceased before the lavas studied (Series I) were erupted. Accepted.

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